

HDOC-29

Aim:

To write a menu-driven C program using switch-case to perform the following operations on arrays:

1. Count even, odd, positive, negative, and zero elements.
2. Search for an element using Linear Search.
3. Merge two arrays.
4. Exit the program.

Task1: Algorithm:

Main Algorithm

1. Start.
2. Declare choice.
3. Display the menu:
 - Count even, odd, positive, negative, and zero elements.
 - Linear search.
 - Merge two arrays.
 - Exit.
4. Read choice from the user.
5. Use switch(choice):
 - If choice = 1, call countElements().
 - If choice = 2, call linearSearch().
 - If choice = 3, call mergeArrays().
 - If choice = 4, exit.
 - Otherwise, display "Invalid choice".
6. Repeat the menu until the user selects 4.
7. Stop.

Algorithm for countElements()

1. Read the size n of the array.
2. Read n array elements.
3. Initialize even, odd, positive, negative, and zero to 0.
4. Traverse the array:
 - If the element is even, increment even; otherwise increment odd.
 - If the element is positive, increment positive.
 - Else if it is negative, increment negative.
 - Otherwise, increment zero.
5. Display all five counts.

Algorithm for linearSearch()

1. Read array size n.
2. Read n elements.
3. Read the element to be searched.
4. Traverse the array from the first element.
5. If the required element is found, display its position.
6. If the complete array is traversed without finding it, display "Element not found".

Algorithm for mergeArrays()

1. Read the size of the first array n1.
2. Read its elements.
3. Read the size of the second array n2.
4. Read its elements.
5. Copy all elements of the first array into the merged array.
6. Copy all elements of the second array after them.
7. Display the merged array.

Task2: Test case table:

Test Case	Choice	Input	Expected Output
1	1	n=6, Array: 2 -3 0 7 -4 6	Even = 4, Odd = 2, Positive = 3, Negative = 2, Zero = 1
2	2	n=5, Array: 10 20 30 40 50, Search = 30	30 found at position 3
3	2	n=5, Array: 10 20 30 40 50, Search = 60	Element not found
4	3	First: 1 2 3, Second: 4 5 6	Merged array: 1 2 3 4 5 6
5	4	Choice = 4	Exiting program...

Task3: C program:

```
#include <stdio.h>
void countElements();
void linearSearch();
void mergeArrays();
int main()
{
    int choice;
    do
    {
        printf("\n----- ARRAY OPERATIONS MENU ----- \n");
        printf("1. Count even, odd, positive, negative and zero elements\n");
```

```

printf("2. Linear Search\n");
printf("3. Merge two arrays\n");
printf("4. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);
switch(choice)
{
    case 1:
        countElements();
        break;
    case 2:
        linearSearch();
        break;
    case 3:
        mergeArrays();
        break;
    case 4:
        printf("Exiting program...\n");
        break;
    default:
        printf("Invalid choice! Please enter 1 to 4.\n");
}
} while(choice != 4);
return 0;
}
/* Function to count even, odd, positive,
negative and zero elements */
void countElements()
{
    int n, i;
    int even = 0, odd = 0;
    int positive = 0, negative = 0, zero = 0;
    printf("Enter size of array: ");
    scanf("%d", &n);
    int a[n];
    printf("Enter %d elements: ", n);
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }
    for(i = 0; i < n; i++)
    {
        if(a[i] % 2 == 0)
            even++;
        else
            odd++;
    }
}

```

```

        if(a[i] > 0)
            positive++;
        else if(a[i] < 0)
            negative++;
        else
            zero++;
    }
    printf("\nEven elements    = %d\n", even);
    printf("Odd elements      = %d\n", odd);
    printf("Positive elements = %d\n", positive);
    printf("Negative elements = %d\n", negative);
    printf("Zero elements    = %d\n", zero);
}
/* Function for linear search */
void linearSearch()
{
    int n, i, key;
    int found = 0;
    printf("Enter size of array: ");
    scanf("%d", &n);
    int a[n];
    printf("Enter %d elements: ", n);
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }
    printf("Enter element to search: ");
    scanf("%d", &key);
    for(i = 0; i < n; i++)
    {
        if(a[i] == key)
        {
            printf("Element %d found at position %d.\n", key, i + 1);
            found = 1;
            break;
        }
    }
    if(found == 0)
    {
        printf("Element not found in the array.\n");
    }
}
/* Function to merge two arrays */
void mergeArrays()
{
    int n1, n2, i;

```

```
printf("Enter size of first array: ");
scanf("%d", &n1);
int a[n1];
printf("Enter %d elements of first array: ", n1);
for(i = 0; i < n1; i++)
{
    scanf("%d", &a[i]);
}
printf("Enter size of second array: ");
scanf("%d", &n2);
int b[n2];
int merged[n1 + n2];
printf("Enter %d elements of second array: ", n2);
for(i = 0; i < n2; i++)
{
    scanf("%d", &b[i]);
}
/* Copy first array */
for(i = 0; i < n1; i++)
{
    merged[i] = a[i];
}
/* Copy second array */
for(i = 0; i < n2; i++)
{
    merged[n1 + i] = b[i];
}
printf("Merged array: ");
for(i = 0; i < n1 + n2; i++)
{
    printf("%d ", merged[i]);
}
printf("\n");
}
```

Task: Output:

```
(kali@kali)-[~/hdoc29.c]  
$ ./a.out
```

—ARRAY OPERATIONS MENU—

1. Count even, odd, positive, negative and zero elements
2. Linear Search
3. Merge two arrays
4. Exit

Enter your choice: 1

Enter size of array: 6

Enter 6 elements: 2

-3

0

7

-4

6

Even elements = 4

Odd elements = 2

Positive elements = 3

Negative elements = 2

Zero elements = 1

—ARRAY OPERATIONS MENU—

1. Count even, odd, positive, negative and zero elements
2. Linear Search
3. Merge two arrays
4. Exit

Enter your choice: 2

Enter size of array: 5

Enter 5 elements: 10

20

30

40

50

Enter elements to search: 30

Element 30 found at position 3.

—ARRAY OPERATIONS MENU—

1. Count even, odd, positive, negative and zero elements
2. Linear Search
3. Merge two arrays
4. Exit

Enter your choice: 2

Enter size of array: 5

Enter 5 elements: 10

20

30

40

50

Enter elements to search: 60

Element not found in the array.

——ARRAY OPERATIONS MENU——

1. Count even, odd, positive, negative and zero elements
2. Linear Search
3. Merge two arrays
4. Exit

Enter your choice: 3

Enter size of first array:3

Enter 3 elements of first array: 1

2

3

Enter size of second array:3

Enter 3 elements of second array: 4

5

6

Merged array: 123456

——ARRAY OPERATIONS MENU——

1. Count even, odd, positive, negative and zero elements
2. Linear Search
3. Merge two arrays
4. Exit

Enter your choice: 4

Exiting program...